Citrus (Valencia) Fertilizer Programme 2026/27.

Fertilise Smarter, Grow Stronger.

Season 2026/27 programme for a 10.00 ha block of citrus (valencia).

PREPARED FOR	Demo / validation
PREPARED BY	Sapling Fertilizer
DATE	25 April 2026
REF	REF-CITRUS-2026

REFERENCE BLOCK — CITRUS VALENCIA · SUNDAYS RIVER VALLEY
REFERENCE · SUNDAYS RIVER VALLEY, EASTERN CAPE

SAPLING · AGRICULTURAL ADVISORY

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SECTION 01

Background.

This programme covers the 2026/27 season for citrus (valencia) across 1 block totalling 10.00 ha. The strategy is built around the soil analysis, the available equipment, and the crop's stage-by-stage demand curve.

PROGRAMME AT A GLANCE

- 1 block totalling 10.00 ha
- 7 applications across 4 blends this season
- 3 targeted foliar sprays addressing soil-availability gaps and stage-peak demand
- 2 open items flagged for tracking through the season

SECTION 02

Soil Report.

Soil analysis for Reference Block Valencia (SRV) (10.00 ha) was reviewed against the crop's nutrient requirements for the season.

Reference Block Valencia (SRV) 10.00 ha

Reference block (published norms) · Mehlich-3 · sampled 1 Mar 2026

Ratios & balance checks

RATIO	READING	TARGET	STATUS	POSITION VS IDEAL
Ca : B	2,600 : 1	≤ 1000 : 1	WARN	
	Soil Ca:B = 2600 exceeds 1000:1 — Ca-induced B deficiency likely, especially at flowering. *Apply foliar B at critical stages (pre-bloom, fruit set)*			
P : Zn	13.9 : 1	≤ 150 : 1	OK	
	Within target — no action.			
Sodium saturation (ESP)	2.5 %	≤ 5 %	OK	
	Within target — no action.			

The loudest signals

> Soil Ca:B = 2600 exceeds 1000:1 — Ca-induced B deficiency likely, especially at flowering.

SECTION 04

Application Walkthrough.

Across the season you'll do 7 fertilizer applications across 1 block (10.0 ha total) of citrus (valencia). Each one is laid out below in calendar order — what to apply, where, how, and why.

JULY 2026	AUGUST 2026	SEPTEMBER 2026	OCTOBER 2026
2 applications	2 applications	1 application	1 application

NOVEMBER 2026
1 application

A Group A

1 block · 10.0 ha · 7 applications
Reference Block Valencia (SRV)

Mid-July 2026

2 PARALLEL PASSES

Two parallel passes on the same day, each via the method that suits its nutrient profile. The two passes together give the trees what they need at this stage.

A1 PASS 1 OF 2 FERTIGATION

Inject 15:0:0 (16) + Ca 19.0% at 58 kg/ha

HOW: Inject through the drip irrigation line during a normal watering cycle.

TOTAL THIS PASS: 580 kg (10.0 ha × 58 kg/ha)

PER HA THIS PASS DELIVERS

N 18.0 kg | Ca 22.1 kg

WHY THIS PASS

Trees are at early-season re-growth — building roots, shoots, and the season's canopy, so this pass leads with calcium and nitrogen. Delivered through the irrigation system so it reaches the active root zone without an extra ground operation.

Mid-August 2026

2 PARALLEL PASSES

Two parallel passes on the same day, each via the method that suits its nutrient profile. The two passes together give the trees what they need at this stage.

A2 PASS 2 OF 2

Spread 6:4:15 (9) + Ca 2.3% + Mg 0.9% + S 0.8% at 274 kg/ha

HOW: Spread evenly under the canopy with a granular spreader.

TOTAL THIS PASS: 2 742 kg (10.0 ha × 274 kg/ha)

A3 PASS 1 OF 2 FERTIGATION

PER HA THIS PASS DELIVERS

A4 PASS 2 OF 2 DRY BROADCAST

Inject 15:0:0 (16) + Ca 19.0% at 58 kg/ha

A5

Mid-September 2026

Week 11 of season

FERTIGATION

Inject 13:3:15 (30) + Ca 9.7% + Mg 1.1% + S 1.5% at 124 kg/ha

HOW	Inject through the drip irrigation line during a normal watering cycle.	TOTAL THIS PASS	1 239 kg of fertilizer (10.0 ha × 124 kg/ha)
STAGE	Vegetative — *leaf and shoot growth, building this year's canopy*		

WHY THIS ONE

Trees are at leaf and shoot growth, building this year's canopy, so this pass leads with potassium and nitrogen. Delivered through the irrigation system so it reaches the active root zone without an extra ground operation.

PER HECTARE THIS PASS DELIVERS

N 31.5 kg | P 7.0 kg | K 37.0 kg | Ca 24.0 kg | Mg 2.9 kg | S 3.8 kg

A6

Mid-October 2026

Week 16 of season

FERTIGATION

Inject 13:3:15 (30) + Ca 9.7% + Mg 1.1% + S 1.5% at 124 kg/ha

HOW	Inject through the drip irrigation line during a normal watering cycle.	TOTAL THIS PASS	1 239 kg of fertilizer (10.0 ha × 124 kg/ha)
STAGE	Vegetative — *leaf and shoot growth, building this year's canopy*		

WHY THIS ONE

Trees are at leaf and shoot growth, building this year's canopy, so this pass leads with potassium and nitrogen. Delivered through the irrigation system so it reaches the active root zone without an extra ground operation.

PER HECTARE THIS PASS DELIVERS

N 31.5 kg | P 7.0 kg | K 37.0 kg | Ca 24.0 kg | Mg 2.9 kg | S 3.8 kg

A7

Mid-November 2026

Week 20 of season

FERTIGATION

Inject 9:2:15 (36) + Ca 7.4% + Mg 1.0% + S 1.3% at 348 kg/ha

HOW	Inject through the drip irrigation line during a normal watering cycle.	TOTAL THIS PASS	3 480 kg of fertilizer (10.0 ha × 348 kg/ha)
STAGE	Reproductive — *flowering and fruit-set — pushing flower spikes and starting nuts*		

WHY THIS ONE

Trees are at flowering and fruit-set — pushing flower spikes and starting nuts, so this pass leads with potassium and nitrogen. Delivered through the irrigation system so it reaches the active root zone without an extra ground operation.

PER HECTARE THIS PASS DELIVERS

N 43.8 kg | P 8.4 kg | K 74.0 kg | Ca 25.8 kg | Mg 3.5 kg | S 4.5 kg

SECTION 05

Foliar Sprays.

Targeted foliar applications addressing soil-availability gaps and stage-peak demand windows.

Group A

3 foliar events

WEEK	BLOCK	STAGE	ANALYSIS	RATE	REASON
3	Reference Block Valencia (SRV)	Pre-bloom / bud break	20.5% B	0.1% spray	Annual B foliar at bud break for flower quality + fruit set. B xylem-mobile only — timing matters.
3	Reference Block Valencia (SRV)	Pre-bloom / bud break	36% Zn	0.5% spray	Zn foliar at bud break — new leaf tissue expansion + enzyme activation. Soil Zn typically unavailable on alkaline SA citrus soils.
6	Reference Block Valencia (SRV)	Vegetative I	20.5% B	1.5 kg	Soil Ca:B = 2600 exceeds 1000:1 — Ca-induced B deficiency likely, especially at flowering.

SECTION 06

Season Totals.

Total applied this season per hectare, averaged across blocks. Drives shopping list and shed planning.

NUTRIENT	APPLIED PER HA
Nitrogen	~165 kg
Phosphorus	~38 kg
Potassium	~208 kg
Calcium	~143 kg
Magnesium	~19 kg
Sulphur	~21 kg

SECTION 07

Year 2 and Beyond.

Multi-year trajectory for this perennial programme.

THIS SEASON

Soil-side corrections in motion

- Pre-season lime / gypsum / dolomitic actions move soil chemistry in the right direction across Year 1, but most won't fully reflect in leaf or kernel until Year 2.
- In-season programme delivers the season's nutrient targets while the soil-side corrections work in the background.

YEAR 2 OUTLOOK

Where the soil should be

- Soil re-test mid-Year-2 to confirm pH / Ca / Mg movement from this season's amendments.
- Per-block rates revisited based on the re-test — blocks still trending wrong direction get a second pass; blocks that hit target shift to maintenance rates.
- Leaf analysis at flowering becomes the primary feedback loop once soil corrections are settled.

YEAR 3+

Maintenance trajectory

- Soil corrections largely complete — annual maintenance lime / gypsum + leaf-driven foliar plan.
- Programme stabilises around the orchard's productive rhythm; biggest variations come from yield carry-over and seasonal weather rather than soil-chemistry shifts.

SECTION 08

Items We're Carrying Into The Season.

Open questions, watchpoints, and defaults — grouped by when they need attention. Replaces the old split between "risk flags", "open items", and "assumptions"; one severity-banded list with cross-bucket duplicates removed.

Watch through the season

Monitor these as the season progresses. Catch them on the next leaf or soil test, or on an in-season check-in.

1. **Irrigation water test (EC, pH, Ca, Mg, Na, HCO_3, Cl).** Required before first Part A injection. Dolomite-country water can produce scale when calcium-stream stock is injected. Borehole water EC + bicarb level affect drip-line pH. *Risk of emitter blockage from $CaCO_3$ precipitate*
2. **Mid-season leaf tissue analysis at tillering and at bulb-init/flowering.** Two natural decision points where the programme can still be steered. Catches deficiencies the soil test missed and validates actual uptake against plan. *Mid-season corrections become reactive, not anticipatory*

END OF PROGRAMME

Ref · REF-CITRUS-2026